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Tool Used: Power BI Desktop

1. Introduction

Human Resource (HR) Analytics helps organizations analyze employee data to make informed business decisions. One of the most important HR metrics is employee attrition, as high attrition can increase recruitment costs, reduce productivity, and affect organizational growth.

This project focuses on creating an interactive HR Analytics Dashboard using Power BI. The dashboard provides a visual representation of employee data and helps identify patterns in attrition across different departments, job roles, age groups, education fields, and salary levels.

The dashboard enables users to explore the data interactively through filters and visualizations, making it easier to understand workforce trends.

2. Project Objective

The objective of this project is to develop an interactive dashboard that provides meaningful insights into employee attrition.

The dashboard aims to:

- Analyze employee attrition.
 - Monitor important HR KPIs.
 - Identify departments with high employee turnover.
 - Understand attrition based on age, salary, education, and job role.
 - Support better HR decision-making through data visualization.
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3. Tools Used

The following tool was used to complete this project:

Tool	Purpose
Power BI Desktop	Dashboard creation and data visualization

4. Dashboard KPIs

The dashboard displays the following key performance indicators:

KPI	Value
Total Employees	1470
Total Attrition	237
Attrition Rate	16.1%
Average Age	37 Years
Average Salary	6.5K
Average Years at Company	7 Years

These KPIs provide a quick overview of the organization's workforce.

5. Dashboard Visualizations

The dashboard includes several interactive visualizations that help analyze employee attrition.

Employee Count

Displays the total number of employees available in the dataset.

Attrition

Shows the total number of employees who have left the organization.

Attrition Rate

Displays the percentage of employees who left the company.

Average Age

Represents the average age of employees.

Average Salary

Shows the average monthly salary of employees.

Average Years at Company

Displays the average number of years employees have worked in the organization.

Attrition by Education

A donut chart showing attrition based on employees' education fields.

It helps identify which educational backgrounds experience higher attrition.

Attrition by Age

A bar chart displaying attrition across different age groups.

This visualization highlights the age groups with the highest employee turnover.

Attrition by Gender

Displays employee attrition based on gender.

It allows comparison between male and female attrition.

Attrition by Salary Slab

Shows employee attrition across different salary ranges.

This helps understand whether salary influences employee turnover.

Attrition by Years at Company

A line chart illustrating attrition based on employees' years of service.

It identifies the stages of employment where attrition is highest.

Attrition by Job Role

A horizontal bar chart ranking job roles according to employee attrition.

This helps identify positions with the highest turnover.

Department Filter

The dashboard includes a department slicer with:

- Human Resources

- Research & Development
- Sales

Selecting any department automatically updates all visualizations on the dashboard.



6. Key Insights

Based on the dashboard analysis, the following observations can be made:

- The organization has **1,470 employees**.
- The overall employee attrition rate is **16.1%**.
- Employees aged **26–35 years** have the highest attrition.
- Employees earning **up to 5K** show the highest turnover.
- Laboratory Technicians experience the highest attrition among all job roles.
- Most employees leave during the early years of their employment.
- Male employees account for a larger share of attrition.
- Life Sciences is the education field with the highest attrition.

7. Dashboard Features

The dashboard offers several useful features:

- Interactive KPI cards

- Department-wise filtering
 - Cross-filtering between visuals
 - Easy-to-understand charts
 - Clean and organized layout
 - Business-focused insights
 - User-friendly interface
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8. Business Benefits

This dashboard helps HR teams:

- Monitor employee attrition.
 - Identify departments with higher turnover.
 - Analyze workforce trends.
 - Support employee retention strategies.
 - Make faster and more informed decisions using visual insights.
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9. Conclusion

The HR Analytics Dashboard demonstrates how Power BI can transform employee data into meaningful and interactive visualizations. By presenting key HR metrics and attrition trends in a clear and organized manner, the dashboard enables users to quickly understand workforce patterns and identify areas that require attention.

This project strengthened my practical knowledge of Power BI, dashboard design, data visualization, KPI creation, and interactive reporting. It also improved my ability to present business data in a way that supports informed decision-making.

10. Skills Demonstrated

- Power BI Dashboard Development
- Data Visualization
- KPI Design
- Interactive Dashboard Creation
- Dashboard Layout and Formatting
- Business Intelligence Reporting

- HR Data Analysis
- Data Storytelling